

Quantifying Uncertainty in Porous Grain Boundaries

Eero Yarborough, Jarrah Rasmussen, Xiulan Castellanos, Sten Quiroga, Imogen Kowalczyk

Meridian Institute of Technology

Section: Materials
Published: 2022-06-20
DOI: 10.5281/mjas.11.2.06
Keywords: cross-validation; parameter sweep; calibration

Abstract

We examine the behaviour of polymer blends in cryogenic service. The measurements reported here were collected over twelve independent trials under controlled conditions. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study.

1. Introduction

We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Residual error grows approximately linearly with the number of coupled parameters. Calibration drift over the trial period remained below the threshold set out in the protocol. Neither of the alternative formulations improved on the simple additive model in our setting. The measurements reported here were collected over twelve independent trials under controlled conditions. We release the processing scripts and intermediate artefacts to support independent verification. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. The measurements reported here were collected over twelve independent trials under controlled conditions. The reduced model retains most of the explanatory power at a fraction of the computational cost. Taken together, these findings argue for treating the two regimes separately in future analyses. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Sensitivity to the smoothing window proved modest across the range we examined. Residual error grows approximately linearly with the number of coupled parameters. Longer observation windows narrow the confidence interval but do not shift the central estimate.

Replicate agreement was high, suggesting the protocol is robust to small handling differences. Residual error grows approximately linearly with the number of coupled parameters. Neither of the alternative formulations improved on the simple additive model in our setting. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Calibration drift over the trial period remained below the threshold set out in the protocol. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity.

Sensitivity to the smoothing window proved modest across the range we examined. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Taken together, these findings

argue for treating the two regimes separately in future analyses. Calibration drift over the trial period remained below the threshold set out in the protocol. We release the processing scripts and intermediate artefacts to support independent verification. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. The measurements reported here were collected over twelve independent trials under controlled conditions.

2. Related Work

Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Calibration drift over the trial period remained below the threshold set out in the protocol. Taken together, these findings argue for treating the two regimes separately in future analyses. We release the processing scripts and intermediate artefacts to support independent verification.

Discrepancies at the boundary are consistent with the simplifying assumptions noted above. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Residual error grows approximately linearly with the number of coupled parameters. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Neither of the alternative formulations improved on the simple additive model in our setting.

Discrepancies at the boundary are consistent with the simplifying assumptions noted above. We release the processing scripts and intermediate artefacts to support independent verification. Residual error grows approximately linearly with the number of coupled parameters. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Calibration drift over the trial period remained below the threshold set out in the protocol. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Sensitivity to the smoothing window proved modest across the range we examined.

Taken together, these findings argue for treating the two regimes separately in future analyses. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Residual error grows approximately linearly with the number of coupled parameters. The reduced model retains most of the explanatory power at a fraction of the computational cost. Neither of the alternative formulations improved on the simple additive model in our setting. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study.

Neither of the alternative formulations improved on the simple additive model in our setting. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Calibration drift over the trial period remained below the threshold set out in the protocol. Residual error grows approximately linearly with the number of coupled parameters.

3. Materials and Methods

The measurements reported here were collected over twelve independent trials under controlled conditions. Taken together, these findings argue for treating the two regimes separately in future analyses. Sensitivity to the smoothing window proved modest across the range we examined. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Residual error grows approximately linearly with the number of coupled parameters. Longer observation windows narrow the confidence interval but do not shift the central estimate.

Taken together, these findings argue for treating the two regimes separately in future analyses. Residual error grows approximately linearly with the number of coupled parameters. The reduced model retains most of the explanatory power at a fraction of the computational cost. We release the processing scripts and intermediate artefacts to support independent verification.

4. Experimental Setup

Neither of the alternative formulations improved on the simple additive model in our setting. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. We release the processing scripts and intermediate artefacts to support independent verification. Taken together, these findings argue for treating the two regimes separately in future analyses. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Calibration drift over the trial period remained below the threshold set out in the protocol. Residual error grows approximately linearly with the number of coupled parameters.

Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Sensitivity to the smoothing window proved modest across the range we examined. Longer observation windows narrow the confidence interval but do not shift the central estimate. We release the processing scripts and intermediate artefacts to support independent verification. The measurements reported here were collected over twelve independent trials under controlled conditions. Taken together, these findings argue for treating the two regimes separately in future analyses.

5. Results

Longer observation windows narrow the confidence interval but do not shift the central estimate. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. The reduced model retains most of the explanatory power at a fraction of the computational cost. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. The measurements reported here were collected over twelve independent trials under controlled conditions. Residual error grows approximately linearly with the number of coupled parameters. Sensitivity to the smoothing window proved modest across the range we examined. Calibration drift over the trial period remained below the threshold set out in the protocol.

The reduced model retains most of the explanatory power at a fraction of the computational cost. The measurements reported here were collected over twelve independent trials under controlled conditions. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Taken together, these findings argue for treating the two regimes separately in future analyses. Longer observation windows narrow the confidence interval but do not shift the central estimate. Sensitivity to the smoothing window proved modest across the range we examined. We observe a pronounced inflection once the load exceeds

roughly sixty percent of nominal capacity. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance.

We release the processing scripts and intermediate artefacts to support independent verification. Longer observation windows narrow the confidence interval but do not shift the central estimate. Neither of the alternative formulations improved on the simple additive model in our setting. The measurements reported here were collected over twelve independent trials under controlled conditions. Sensitivity to the smoothing window proved modest across the range we examined. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Taken together, these findings argue for treating the two regimes separately in future analyses. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Replicate agreement was high, suggesting the protocol is robust to small handling differences.

6. Discussion

Residual error grows approximately linearly with the number of coupled parameters. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Neither of the alternative formulations improved on the simple additive model in our setting. The reduced model retains most of the explanatory power at a fraction of the computational cost. We release the processing scripts and intermediate artefacts to support independent verification. The measurements reported here were collected over twelve independent trials under controlled conditions. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study.

We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Replicate agreement was high, suggesting the protocol is robust to small handling differences. Taken together, these findings argue for treating the two regimes separately in future analyses. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. The reduced model retains most of the explanatory power at a fraction of the computational cost. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Longer observation windows narrow the confidence interval but do not shift the central estimate. Calibration drift over the trial period remained below the threshold set out in the protocol. Residual error grows approximately linearly with the number of coupled parameters.

The measurements reported here were collected over twelve independent trials under controlled conditions. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Residual error grows approximately linearly with the number of coupled parameters. Longer observation windows narrow the confidence interval but do not shift the central estimate. Replicate agreement was high, suggesting the protocol is robust to small handling differences. The reduced model retains most of the explanatory power at a fraction of the computational cost. We release the processing scripts and intermediate artefacts to support independent verification.

Replicate agreement was high, suggesting the protocol is robust to small handling differences. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Sensitivity to the smoothing window proved modest across the range we examined. The measurements reported here were collected over twelve independent trials under controlled conditions.

We release the processing scripts and intermediate artefacts to support independent verification. Taken

together, these findings argue for treating the two regimes separately in future analyses. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. The measurements reported here were collected over twelve independent trials under controlled conditions. Sensitivity to the smoothing window proved modest across the range we examined.

The reduced model retains most of the explanatory power at a fraction of the computational cost. Residual error grows approximately linearly with the number of coupled parameters. The measurements reported here were collected over twelve independent trials under controlled conditions. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Replicate agreement was high, suggesting the protocol is robust to small handling differences.